UNITED ARAB EMIRATES MINISTRY OF EDUCATION
الإمارات العربية المتحدة وزارة التربية والتعليم

Al Khalidiya Boys School – Cycle 3

School code 10048 | Dibba Al Hisn | Sharjah Eastern

Grade 9 Physics Final Exam Practice

Term 1 2026–2027 • Set B • With Answers

100 marks | 150 minutes | Calculator allowed

Answer all 20 questions. Choose one option for each MCQ. Show working, units and labelled diagrams for Q17–20. Use g = 9.8 m/s² when needed.

Section A Multiple Choice 60 marks

1. A scientist obtains data that contradict a hypothesis. What is the best next action? [4 marks]

A. State that evidence cannot challenge a hypothesis
B. Remove the conflicting data without a reason
C. Declare the experiment unnecessary
D. Revise the hypothesis and test again

Reference: pp. 4–6; p. 8, Check Your Progress 1–3 | **Answer D**

2. Which is a scientific model? [4 marks]

A. An untestable statement about beauty
B. A vote on the best-looking vehicle
C. An instrument reading with no interpretation
D. A mathematical relation used to predict motion

Reference: pp. 6–7 | **Answer D**

3. An evidence-based explanation is revised after new observations. What does this show? [4 marks]

A. Scientific theories can develop when evidence changes
B. The explanation was never scientific
C. Scientific laws are matters of opinion
D. All previous measurements must be false

Reference: p. 7 | **Answer A**

4. Which conversion is correct? [4 marks]

A. 4.2 km = 4.2 × 10^3 m
B. 4.2 km = 420 m
C. 4.2 km = 4.2 × 10^{-3} m
D. 4.2 km = 42 m

Reference: pp. 10–11 | **Answer A**

5. Find 13.78 g ÷ 11.3 mL with the correct significant digits. [4 marks]

A. 1.21947 g/mL
B. 1 g/mL
C. 1.22 g/mL
D. 1.2 g/mL

Reference: p. 12, Check Your Progress 8(d) | **Answer C**

6. A lap time is reported as 65.414 s using a stopwatch that reads to 0.1 s. Which report is appropriate? [4 marks]

A. 65.4 s
B. 65.4140 s
C. 65.41 s
D. 65 s

Reference: p. 16, Check Your Progress 17 | **Answer A**

7. On a spring length–mass graph, what does a smaller slope mean? [4 marks]

A. Less extension per gram of added mass
B. More extension for the same added mass
C. A greater initial length in every case
D. No relationship between mass and length

Reference: p. 22, Check Your Progress 23 | **Answer A**

8. When is a particle model most appropriate? [2 marks]

A. Only when an object is at rest
B. When the object is small compared with the distance moved
C. Only when an object has zero mass
D. When the detailed rotation of every part is required

Reference: pp. 30–31 | **Answer B**

9. Which physical quantity is a scalar? [4 marks]

A. Acceleration
B. Displacement
C. Pressure
D. Force

Reference: p. 33 | **Answer C**

10. Which particle pattern corresponds to a straight position–time line with positive slope? [4 marks]

A. Points with increasing gaps moving positively
B. Points with decreasing gaps moving negatively
C. Points all at one position
D. Equally spaced points moving in the positive direction

Reference: pp. 37–38; p. 39, Practice Problem 11 | **Answer D**

11. Two straight position–time lines intersect. What does the intersection mean? [4 marks]

A. Both objects must have zero speed
B. The objects have equal masses
C. The two objects are at the same position at the same time
D. The objects have equal accelerations only

Reference: p. 40, Example Problem 2 | **Answer C**

12. An object covers equal displacements in equal time intervals along a straight line. Its motion is: [4 marks]

A. Always accelerated
B. Impossible to model
C. Necessarily at rest
D. Uniform

Reference: pp. 43–44; 54–56 | **Answer D**

13. An object moves east but has acceleration west. What happens to its speed while it still moves east? [4 marks]

A. It stays constant
B. It decreases
C. It is already zero
D. It increases

Reference: p. 56 | **Answer B**

14. The same golf ball starts at +2.0 m/s with $a = -0.50\ m/s^2$. What is its velocity after 6.0 s? [4 marks]

A. +5.0 m/s
B. +1.0 m/s
C. −1.0 m/s
D. −3.0 m/s

Reference: p. 65, Practice Problem 16(b) | **Answer C**

15. A car increases speed from 15 m/s to 25 m/s over 125 m. What is its constant acceleration? [4 marks]

A. 5.0 m/s^2
B. 1.6 m/s^2
C. 0.08 m/s^2
D. 3.2 m/s^2

Reference: p. 68, Practice Problem 25 | **Answer B**

16. Two objects with different masses are released together in a vacuum. They fall with: [2 marks]

A. Zero acceleration
B. The same acceleration
C. Acceleration proportional to mass
D. The lighter object's acceleration always larger

Reference: pp. 71–74 | **Answer B**

Set B Section B Free Response

17. A graduated cylinder has scale divisions of 1 mL. A beaker has divisions of 50 mL. Use the textbook rule that reading uncertainty is half the smallest division. [10 marks]

Reference: p. 14, Precision Versus Accuracy; pp. 17–18, Spring Investigation

(a) State the reading uncertainty of each instrument. [2]

Answer Cylinder: ±0.5 mL [1]; beaker: ±25 mL [1].

(b) Which instrument gives a more precise volume? Explain. [2]

Answer Graduated cylinder [1]; smaller divisions and smaller reading uncertainty [1].

(c) Convert both uncertainties into SI volume units. Use $1\ mL = 10^{-6}\ m^3$. [2]

Answer $\pm 5.0 \times 10^{-7}\ m^3$ [1]; $\pm 2.5 \times 10^{-5}\ m^3$ [1].

(d) In a spring investigation, added mass is varied and spring length is measured. Identify the independent and dependent variables, and name a suitable instrument for each. [4]

Answer Independent: added mass [1], balance [1]. Dependent: spring length [1], ruler [1].

18. Take east as positive. A ship starts 25 km east of A and travels west at 35 km/h for 1.2 h. Rohith runs east, slowing from 3.5 m/s to 0.75 m/s in 10.0 s. [10 marks]

Reference: p. 47, Practice Problem 34; p. 61, Practice Problem 9

(a) Find the ship's displacement. [2]

Answer $\Delta x = (-35)(1.2)$ [1] = −42 km, west [1].

(b) Find its final position relative to A. [2]

Answer $x_f = 25 - 42$ [1] = −17 km, or 17 km west of A [1].

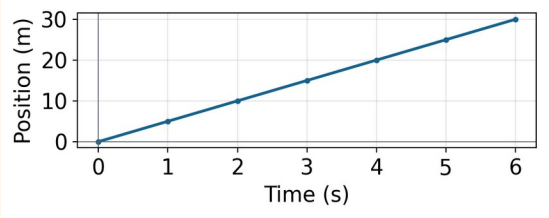
(c) State the ship's average speed and average velocity during this trip. [2]

Answer Average speed 35 km/h [1]; average velocity −35 km/h, west [1].

(d) Calculate Rohith's average acceleration. Is it in the direction of motion? [4]

Answer $a = (0.75 - 3.5)/10$ [1] = −0.275 m/s² [1], west [1]. It opposes his eastward velocity, so he slows [1].

19. A runner's straight position–time line passes through (0.0 s, 0.0 m) and (6.0 s, 30.0 m). Use the graph. [10 marks]



Reference: p. 39, Example Problem 1, Figure 11; pp. 32–39

(a) Calculate the runner's constant velocity. [2]

Answer v = slope = (30 − 0)/(6 − 0) [1] = +5.0 m/s [1].

(b) Find when the runner reaches 12.0 m and the position at 4.5 s. [3]

Answer x = vt [1]; t = 12/5 = 2.4 s [1]; x = 5(4.5) = 22.5 m [1].

(c) Calculate displacement and distance between t = 2.0 s and t = 5.0 s. [3]

Answer x(2) = 10 m and x(5) = 25 m [1]; Δx = +15 m [1]; distance = 15 m [1].

(d) Explain why the runner is not accelerating and how the direction is shown. [2]

Answer Straight line has constant slope and hence constant velocity [1]; positive slope indicates positive-direction motion [1].

20. A car starts from rest and accelerates uniformly westward to 25 m/s in 12 s. A second car travels at constant velocity and covers the same displacement in the same time. Take west as positive. [10 marks]

Reference: p. 66, Practice Problem 22

(a) Draw a labelled velocity–time graph for the accelerating car. [3]

Answer Time (s) and velocity (m/s) axes [1]; start (0,0) [1]; straight line to (12,25) [1].

(b) Calculate the first car's acceleration. [2]

Answer $a = (25 - 0)/12$ [1] = +2.08 m/s², west [1].

(c) Calculate its displacement from the graph area. [3]

Answer Triangle area ½ × 12 × 25 [2 for method and substitution] = +150 m, west [1].

(d) Find the constant velocity of the second car. [2]

Answer $v = \Delta x/t = 150/12$ [1] = +12.5 m/s, west [1].

